

Solution to Ramanujan Challenge Problem 2.3: The Sum $\pi + e$ as an Apéry Limit

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Abstract

We prove that the order-four recurrence of Problem 2.3 converges to $\pi + e$. The mechanism is that the challenge operator is a *tensor product*: order $4 = 2 \times 2$. It annihilates every product $X_{n+2}Y_{n+3}$ of a solution X of Lambert's continuant recurrence (whose ratio produces $\pi/4$) with a solution Y of the derangement recurrence (whose ratio produces e). The key observation is that $m!$ satisfies the *same* recurrence as the derangement numbers D_m ; this is why π and e appear together, and additively. The challenge's own p_n, q_n are then identified with explicit tensor solutions, and the ratio splits *exactly*, with no asymptotic analysis:

$$\frac{p_n}{q_n} = 4 \frac{B_{n+2}}{A_{n+2}} + \frac{(n+3)!}{D_{n+3}}.$$

Everything above is proved here and machine-checked in Lean 4. The single classical input we do not reprove is the value of Lambert's continued fraction, $B_m/A_m \rightarrow \pi/4$; it is isolated as an explicit hypothesis of the formal statement, so that the dependency is visible rather than hidden.

1 The problem

For $n \geq 1$ the challenge poses the recurrence

$$\begin{aligned} 0 = & (-n^3 + 2n^2 + 7n + 3) u_n \\ & + (n+2)(2n^4 + n^3 - 26n^2 - 48n - 19) u_{n-1} \\ & + (n+2)(n^6 + 9n^5 + 8n^4 - 87n^3 - 249n^2 - 234n - 68) u_{n-2} \\ & + (n+1)^2(n+2)(2n^5 + 3n^4 - 13n^3 - 21n^2 + 4) u_{n-3} \\ & - n^3(n+1)^2(n+2)(n^3 + n^2 - 8n - 11) u_{n-4}, \end{aligned} \tag{1}$$

and asks for $\lim_{n \rightarrow \infty} p_n/q_n = \pi + e$, where p, q are the solutions with

$$(p_{-3}, p_{-2}, p_{-1}, p_0) = (1, 1, 20, 296), \quad (q_{-3}, q_{-2}, q_{-1}, q_0) = (1, 0, 4, 48). \tag{2}$$

Write $c_0(n), \dots, c_4(n)$ for the five coefficient polynomials in (1), in the order displayed, so that (1) reads $\sum_{j=0}^4 c_j(n) u_{n-j} = 0$.

2 Two order-two systems

Definition 2.1. The Lambert system $\mathcal{L}$ is the recurrence

$$X_m = (2m+1)X_{m-1} + m^2X_{m-2} \quad (m \geq 1), \quad (3)$$

and the derangement system $\mathcal{D}$ is

$$Y_m = (m-1)(Y_{m-1} + Y_{m-2}) \quad (m \geq 2). \quad (4)$$

Distinguished solutions: A_m, B_m of $\mathcal{L}$ with $A_{-1} = A_0 = 1$ and $B_{-1} = 0, B_0 = 1$; and D_m (the derangement numbers) of $\mathcal{D}$ with $D_0 = 1, D_1 = 0$.

Thus $A_1 = 4, A_2 = 24, A_3 = 204; B_1 = 3, B_2 = 19, B_3 = 160$; and $D_2 = 1, D_3 = 2, D_4 = 9$.

Lemma 2.2 (The factorial is a second solution of $\mathcal{D}$). $F_m := m!$ satisfies (4).

Proof. $(m-1)((m-1)! + (m-2)!) = (m-2)!(m-1)((m-1)+1) = (m-2)!(m-1)m = m!$. $\square$

Lemma 2.2 is the structural reason the constant e occurs in this problem, and the reason it occurs *additively* alongside π : the $\mathcal{D}$ -system carries both D_m and $m!$, and their ratio is the classical approximation to e .

Lemma 2.3 (Casorati determinants). For all $m \geq 0$,

$$A_mB_{m-1} - A_{m-1}B_m = (-1)^{m+1}(m!)^2, \quad D_mF_{m-1} - D_{m-1}F_m = (-1)^m(m-1)! \quad (m \geq 1).$$

In particular $\{A, B\}$ is a basis of the solution space of $\mathcal{L}$ and $\{D, F\}$ is a basis of the solution space of $\mathcal{D}$.

Proof. Write $W_m = A_mB_{m-1} - A_{m-1}B_m$. Substituting (3) for A_m and B_m , the $(2m+1)$ -terms cancel and $W_m = -m^2 W_{m-1}$; with $W_0 = A_0B_{-1} - A_{-1}B_0 = -1$ this telescopes to $W_m = (-1)^{m+1}(m!)^2$. The second identity is the same computation: substituting (4) gives $D_mF_{m-1} - D_{m-1}F_m = -(m-1)(D_{m-1}F_{m-2} - D_{m-2}F_{m-1})$, and the initial value at $m = 1$ is $D_1F_0 - D_0F_1 = -1$. $\square$

3 The tensor theorem

Theorem 3.1 (Tensor annihilation). Let X be any solution of $\mathcal{L}$ and Y any solution of $\mathcal{D}$. Then

$$u_n := X_{n+2}Y_{n+3}$$

satisfies the challenge recurrence (1) for every $n \geq 1$.

Proof. The assertion is an identity in the four free initial values X_{-1}, X_0, Y_0, Y_1 , so it suffices to prove a polynomial identity. Put

$$\mathbf{a}(m) = \begin{pmatrix} 2m+1 & m^2 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{d}(m) = \begin{pmatrix} m-1 & m-1 \\ 1 & 0 \end{pmatrix},$$

so that $(X_m, X_{m-1})^\top = \mathbf{a}(m)(X_{m-1}, X_{m-2})^\top$ and $(Y_m, Y_{m-1})^\top = \mathbf{d}(m)(Y_{m-1}, Y_{m-2})^\top$. Then the vector

$$v_n := (X_{n+2}, X_{n+1})^\top \otimes (Y_{n+3}, Y_{n+2})^\top \in \mathbb{Q}^4$$

obeys $v_n = T(n) v_{n-1}$ with $T(n) = \mathfrak{a}(n+2) \otimes \mathfrak{d}(n+3)$, and $u_n = e_1^\top v_n$ where e_1 is the first standard covector. Hence

$$\sum_{j=0}^4 c_j(n) u_{n-j} = \left(\sum_{j=0}^4 c_j(n) e_1^\top T(n) T(n-1) \cdots T(n-j+1) \right) v_{n-4},$$

the empty product being the identity. As the initial data vary, v_{n-4} ranges over a spanning set of $\mathbb{Q}^4$ (Lemma 2.3 supplies four independent products), so the recurrence holds for all solutions if and only if the covector in parentheses vanishes identically — that is, if and only if four polynomial identities in $\mathbb{Z}[n]$ hold.

Concretely, one eliminates X_{n+2}, X_{n+1}, X_n and $Y_{n+3}, Y_{n+2}, Y_{n+1}$ using (3) and (4) until $\sum_j c_j(n) u_{n-j}$ is a bilinear form in the four products

$$X_{n-2}Y_{n-1}, \quad X_{n-2}Y_n, \quad X_{n-1}Y_{n-1}, \quad X_{n-1}Y_n,$$

whose four coefficients are polynomials in n . All four are identically zero.

This is a finite computation in $\mathbb{Z}[n]$, and it has been carried out three independent ways: (i) in Lean 4, where after substitution it is discharged by the `ring` normaliser and is therefore kernel-checked (`RamanujanChallenge.P23.tensor_rec`); (ii) symbolically over $\mathbb{Q}[n]$ with four free initial values; and (iii) in exact integer arithmetic for the concrete sequences. See §6 and §7. $\square$

Corollary 3.2. *The solution space of (1) over $\mathbb{Q}$ is exactly four dimensional and is spanned by*

$$\{A_{n+2}D_{n+3}, A_{n+2}F_{n+3}, B_{n+2}D_{n+3}, B_{n+2}F_{n+3}\}.$$

Thus the challenge operator is the tensor product of $\mathcal{L}$ and $\mathcal{D}$, not merely annihilated by it.

Proof. By Lemma 2.3 the four listed sequences are linearly independent: their value matrix at two consecutive indices is the Kronecker product of two invertible 2×2 matrices. By Theorem 3.1 they solve (1). Since $c_0(n) \neq 0$ for $n \geq 1$ (Lemma 4.1 below), the solution space of (1) has dimension exactly four. $\square$

4 Identification of the challenge sequences

Lemma 4.1. $c_0(n) = -n^3 + 2n^2 + 7n + 3 \neq 0$ for every integer $n \geq 1$; consequently a solution of (1) is uniquely determined by $(u_{-3}, u_{-2}, u_{-1}, u_0)$.

Proof. For $n = 1, 2, 3, 4$ the values are 11, 17, 15, -1 . For $n \geq 5$, $n^3 - 2n^2 - 7n - 3 = n^2(n-2) - 7n - 3 \geq 3n^2 - 7n - 3 > 0$, so $c_0(n) < 0$. As $c_0(n) \neq 0$, equation (1) solves for u_n at each $n \geq 1$. $\square$

Proposition 4.2 (Closed forms). *The challenge sequences of (2) are*

$$q_n = A_{n+2} D_{n+3}, \quad p_n = 4 B_{n+2} D_{n+3} + A_{n+2} (n+3)!. \quad (5)$$

Proof. By Theorem 3.1 the sequence $A_{n+2}D_{n+3}$ solves (1); so do $B_{n+2}D_{n+3}$ and, by Lemma 2.2, $A_{n+2}(n+3)!$. Since (1) is linear, the right-hand sides of (5) are solutions. It remains to check four initial values of each, using $A_{-1} = A_0 = 1, A_1 = 4, A_2 = 24, B_{-1} = 0, B_0 = 1, B_1 = 3, B_2 = 19$ and $D_0 = 1, D_1 = 0, D_2 = 1, D_3 = 2$:

$$\begin{aligned} A_{-1}D_0 &= 1, & A_0D_1 &= 0, & A_1D_2 &= 4, & A_2D_3 &= 48; \\ 4B_{-1}D_0 + A_{-1}0! &= 1, & 4B_0D_1 + A_01! &= 1, & 4B_1D_2 + A_12! &= 20, & 4B_2D_3 + A_23! &= 296. \end{aligned}$$

These are exactly (2). By Lemma 4.1 the solution with prescribed initial data is unique, so (5) holds for all n . $\square$

Corollary 4.3 (Exact splitting). *For every $n \geq -1$,*

$$\frac{p_n}{q_n} = 4 \frac{B_{n+2}}{A_{n+2}} + \frac{(n+3)!}{D_{n+3}}. \quad (6)$$

Proof. Divide (5). The denominators are nonzero: $A_m > 0$ for all m by induction on (3) (both coefficients are positive for $m \geq 1$), and $D_m > 0$ for $m \geq 2$ likewise. $\square$

We stress that (6) is an identity, valid term by term; no asymptotic or Poincaré–Perron argument is used to obtain it. All that remains is to evaluate the two limits separately.

5 The two limits

Lemma 5.1 (The e half). $\lim_{m \rightarrow \infty} \frac{m!}{D_m} = e$.

Proof. Write $S_m := \sum_{k=0}^m (-1)^k / k!$. We claim $D_m = m! S_m$ for all $m \geq 0$. This holds at $m = 0$ ($1 = 1 \cdot 1$) and $m = 1$ ($0 = 1 \cdot (1 - 1)$). Assume it at $m-1$ and $m-2$. Using $(m-1)! = (m-1)(m-2)!$,

$$(m-1)(D_{m-1} + D_{m-2}) = (m-1)[(m-1)! S_{m-1} + (m-2)! S_{m-2}] = (m-1)! [(m-1)S_{m-1} + S_{m-2}].$$

Now $S_{m-2} = S_{m-1} - (-1)^{m-1}/(m-1)!$, so the bracket equals $m S_{m-1} - (-1)^{m-1}/(m-1)!$ and therefore

$$(m-1)(D_{m-1} + D_{m-2}) = m! S_{m-1} - (-1)^{m-1} = m! S_{m-1} + (-1)^m = m! \left(S_{m-1} + \frac{(-1)^m}{m!} \right) = m! S_m,$$

which is D_m . Hence $m!/D_m = S_m^{-1}$. The partial sums S_m converge to $e^{-1} \neq 0$, and $S_m > 0$ for $m \geq 2$ because an alternating series with strictly decreasing terms has all its partial sums from index 2 on trapped between $S_2 = 1/2$ and $S_3 = 1/3$. So the reciprocals converge to e . $\square$

Theorem 5.2 (The π half; Lambert). $\lim_{m \rightarrow \infty} \frac{B_m}{A_m} = \frac{\pi}{4}$.

This is classical, and it is the one ingredient of this solution that we cite rather than reprove. Precisely: A_m and B_m are the continuant sequences (denominators and numerators) of Lambert's continued fraction for $\arctan$ evaluated at $z = 1$,

$$\arctan z = \frac{z}{1 + \frac{z^2}{3 + \frac{4z^2}{5 + \frac{9z^2}{7 + \ddots}}}}, \quad \frac{\pi}{4} = \arctan 1,$$

whose partial numerators are m^2 and partial denominators $2m+1$ — exactly the coefficients of (3). Convergence and the value are due to Lambert [4] and Legendre [5]; modern treatments via Gauss's continued fraction for ratios of ${}_2F_1$ are in Wall [6, §92] and Jones–Thron [2, Ch. 6]. An equivalent form is Brouncker's continued fraction for $4/\pi$, treated in detail by Khrushchev [3].

What we *can* supply unconditionally is the structure of the approximation, which pins down everything except the value of the limit.

Proposition 5.3 (Existence and rate). *The sequence B_m/A_m converges. Indeed, by Lemma 2.3,*

$$\frac{B_m}{A_m} = 1 + \sum_{j=1}^m \frac{(-1)^j (j!)^2}{A_j A_{j-1}}, \quad (7)$$

an alternating series whose terms decrease strictly to 0. Moreover, under the gauge $X_m = m! Z_m$ the recurrence (3) becomes $Z_m = \frac{2m+1}{m} Z_{m-1} + \frac{m}{m-1} Z_{m-2}$, with limiting characteristic polynomial $r^2 - 2r - 1$ and roots $1 \pm \sqrt{2}$. Hence A_m is the dominant solution, the linear form $R_m := A_m \frac{\pi}{4} - B_m$ is the recessive one, and

$$\left| \frac{\pi}{4} - \frac{B_m}{A_m} \right| = \frac{|R_m|}{A_m} = O((3 - 2\sqrt{2})^m), \quad 3 - 2\sqrt{2} = 0.17157\dots$$

Proof. Dividing $W_m = A_m B_{m-1} - A_{m-1} B_m = (-1)^{m+1} (m!)^2$ by $A_m A_{m-1}$ gives $B_{m-1}/A_{m-1} - B_m/A_m = (-1)^{m+1} (m!)^2 / (A_m A_{m-1})$; summing from 1 to m and using $B_0/A_0 = 1$ yields (7). For the terms $t_m := (m!)^2 / (A_m A_{m-1})$, positivity of the coefficients gives $A_m \geq (2m+1)A_{m-1}$ and hence $A_{m+1} \geq [(2m+3)(2m+1) + (m+1)^2]A_{m-1}$, so

$$\frac{t_{m+1}}{t_m} = \frac{(m+1)^2 A_{m-1}}{A_{m+1}} \leq \frac{(m+1)^2}{5m^2 + 10m + 4} \xrightarrow{m \rightarrow \infty} \frac{1}{5}.$$

Thus t_m is strictly decreasing and tends to 0 geometrically, and the alternating series (7) converges. This much is unconditional and elementary.

The final assertion about the rate is the only place where we invoke asymptotic theory. The gauge computation is immediate; since the limiting characteristic roots $1 \pm \sqrt{2}$ have distinct moduli, Poincaré's theorem (see [1, Ch. 8]) applies, giving that every solution has $|Z_m|^{1/m}$ tending to $|1 + \sqrt{2}|$ or $|1 - \sqrt{2}|$, and the linear form R_m is the recessive one. The stated exponent is the ratio $|(1 - \sqrt{2})/(1 + \sqrt{2})| = 3 - 2\sqrt{2}$. Nothing in Theorem 5.4 depends on this paragraph. $\square$

So the content of Theorem 5.2 that we import is precisely the *value* $\pi/4$ of an alternating series that is unconditionally shown here to converge geometrically. Numerically, $|R_m|/A_m$ agrees with $0.221 \cdot (3 - 2\sqrt{2})^m$ to three significant figures for $10 \leq m \leq 20$.

Theorem 5.4 (Main). $\lim_{n \rightarrow \infty} \frac{p_n}{q_n} = \pi + e$.

Proof. Combine the exact splitting (6) with Lemma 5.1 and Theorem 5.2:

$$\frac{p_n}{q_n} = 4 \frac{B_{n+2}}{A_{n+2}} + \frac{(n+3)!}{D_{n+3}} \longrightarrow 4 \cdot \frac{\pi}{4} + e = \pi + e. \quad \square$$

Remark 5.5 (Why the problem looks hard). The four Poincaré behaviours of (1) are the products of the two $\mathcal{L}$ -behaviours (after a factorial gauge, $1 \pm \sqrt{2}$) with the two $\mathcal{D}$ -behaviours. The dominant solution is not the one carrying the answer; p_n and q_n are particular mixtures whose ratio has the two constants sitting in *different* tensor factors. Any attempt to read $\pi + e$ off the order-four recurrence directly must therefore separate a rank-two continuant factor from a rank-two factorial factor, which is what Theorem 3.1 does in closed form.

6 Formal verification

The accompanying Lean 4 development (`lean/`, toolchain `leanprover/lean4:v4.30.0`, Mathlib pinned in `lake-manifest.json`) builds with **0 errors and 0 sorry** and formalises the whole argument above except Theorem 5.2. The principal statements are:

<code>tensor_rec</code>	Theorem 3.1, for arbitrary X, Y
<code>factorial_isDerRec</code>	Lemma 2.2
<code>challengeQ_rec, challengeP_rec</code>	the closed forms solve (1)
<code>challengeQ_zero... challengeP_three</code>	the eight initial values (2)
<code>C0_ne_zero, eq_of_satisfiesRec</code>	Lemma 4.1 and uniqueness
<code>ratio_split</code>	Corollary 4.3
<code>factorial_div_derang_tendsto_exp_one</code>	Lemma 5.1
<code>problem23_pi_add_e</code>	Theorem 5.4

Every one of these depends only on the three standard axioms `propext`, `Classical.choice`, `Quot.sound`; the purely algebraic ones (`tensor_rec`, `challengeQ_rec`, `challengeP_rec`) do not even use `Classical.choice`. There is no `native_decide` anywhere. The e half is obtained from Mathlib’s `numDerangements_tendsto_inv_e` after proving that our `derang` agrees with Mathlib’s `numDerangements`.

Theorem 5.2 appears as an explicit *hypothesis*:

```
theorem problem23_pi_add_e
  (hLambert : Tendsto (fun m => (lambertB m : R) / (lambertA m : R))
    atTop (nhds (Real.pi / 4))) :
  Tendsto (fun m => (challengeP m : R) / (challengeQ m : R))
    atTop (nhds (Real.pi + Real.exp 1))
```

We chose this over introducing an axiom or a `sorry` so that the reliance on the classical continued fraction is legible in the statement itself.

7 Numerical verification

The script `verify.py` performs three checks in exact arithmetic.

1. It solves (1) forward over $\mathbb{Q}$ from the initial data (2) alone and confirms that the result is integral and agrees with the closed forms (5). The first new values are $q_1 = 1836$, $q_2 = 97680$, $p_1 = 10656$, $p_2 = 573344$.
2. It verifies Theorem 3.1 symbolically, with four free initial values, so that no accidental agreement of a particular solution can be mistaken for the general identity.
3. It evaluates p_n/q_n in high precision: at $n = 50$ the agreement with $\pi + e = 5.859874482048838\dots$ is to 40 decimal places, consistent with the geometric rate $(3 - 2\sqrt{2})^n$ of Proposition 5.3.

References

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